

Security+ 4.4 · Security Monitoring

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.4** — Security Monitoring (10:27) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Security monitoring relies on log aggregation (collecting logs from all sources), correlation (linking events across different systems such as login, firewall, and EDR), real-time alerting, and both executive and technical reporting. Logs are the foundation: without logs, there is no forensics. Retention must comply with legal and audit requirements.

High-Yield Core Concepts Tested on Exam

- SIEM aggregates and correlates logs; without logs, none of this is possible
- Correlation example: VPN login + firewall deny + EDR alert = possible attack
- Log aggregation collects data from multiple sources into a central repository
- Alerting provides real-time notification of correlated events
- Reporting serves both executive (summary) and technical (detailed) audiences
- Log retention policies are driven by legal and audit requirements

Exam Tips & Classic Scenarios

- If the question mentions 'combines logs from multiple sources and correlates', the answer is SIEM
- Remember: no logs = no forensics; log retention is critical for incident response

Typical Exam Scenario: A security analyst observes a VPN login event at 8:10, a firewall deny at 8:11, and an EDR block at 8:12. The SIEM correlates these events to identify a potential attack.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.4 — Security Monitoring)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **SIEM:** Security Information and Event Management: Centralized platform performing real-time ingestion, normalization, correlation, and alerting across multi-source security telemetry.
- **EDR:** Endpoint Detection and Response: Agent installed on endpoints providing behavioral telemetry, threat detection, investigation capabilities, and live remote isolation.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. A tool that collects and CORRELATES logs from multiple systems to find threats is:

- A)** Firewall
- B)** SIEM
- C)** Proxy
- D)** Antivirus

2. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SIEM aggregates + correlates; without logs none of this exists'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: SIEM aggregates + correlates; without logs none of this exists...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

3. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'Correlation: VPN login + firewall denied + EDR alert = possible attack'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Correlation: VPN login + firewall denied + EDR alert = possible attack
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'combines logs from multiple sources and correlates'=SIEM. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'combines logs from multiple sources and correlates'=SIEM...

5. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'VPN login event 8:10, firewall denied 8:11, EDR blocks 8:12' = SIEM correlation. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'VPN login event 8:10, firewall denied 8:11, EDR blocks 8:12' = SIEM correlation

6. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SIEM aggregates + correlates; without logs none of this exists'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: SIEM aggregates + correlates; without logs none of this exists...

7. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'Correlation: VPN login + firewall denied + EDR alert = possible attack'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Correlation: VPN login + firewall denied + EDR alert = possible attack...
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'combines logs from multiple sources and correlates'=SIEM'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'combines logs from multiple sources and correlates'=SIEM...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'VPN login event at 8:10, firewall denied at 8:11, EDR blocks at 8:12' = SIEM correlation. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'VPN login event at 8:10, firewall denied at 8:11, EDR blocks at 8:12' = SIEM correlation
- D)** Allow shared access among administrators with generic credentials

10. Corporate scenario (Security Monitoring): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.4), an analyst evaluates the following technical requirement: 'SIEM aggregates + correlates; without logs none of this exists'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: SIEM aggregates + correlates; without logs none of this exists...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — SIEM.

✓ **Why it's correct:** The correct answer is SIEM (Security Information and Event Management). In the CompTIA Security+ SY0-701 Security Architecture and Operations domain, SIEM is the central platform that aggregates, normalizes, and correlates logs and events from multiple heterogeneous sources (firewalls, servers, endpoints, applications, IDS/IPS, routers, etc.). Correlation is the core of SIEM: it applies rules, signatures, and increasingly behavioral analytics (UEBA) to identify patterns that would not be detected in isolation. For example, a failed login on a Linux server followed by a successful access to a file share and an unusual outbound connection to an external IP can be correlated as possible lateral movement and exfiltration. SIEM also provides dashboards, real-time alerts, log retention for compliance, and support for forensic investigations. It is the tool specifically designed for the function described in the question: collecting and correlating logs from multiple systems to find threats.

✗ **Why the others are wrong:**

- *Firewall:* A firewall is a network security device that filters traffic based on access control rules (IPs, ports, protocols, connection states). It can generate logs of allowed and denied connections, but its primary function is to enforce perimeter or segmentation security policies, not to collect and correlate logs from other systems. A firewall does not aggregate logs from servers, endpoints, or applications; it only records its own events. Therefore, it is incapable of performing the multi-source correlation necessary to detect complex threats that span different layers of the infrastructure.
- *Proxy:* A proxy acts as an intermediary for network requests, typically at the application layer (HTTP/HTTPS), providing access control, caching, content filtering, and anonymization. It generates web traffic logs, but these logs are limited to the scope of the traffic that passes through it. A proxy does not collect logs from operating systems, databases, authentication, or other security devices. Its correlation capability is restricted to its own domain and does not replace a SIEM, which integrates and correlates events from the entire infrastructure.
- *Antivirus:* Antivirus is an endpoint solution that detects and removes malware based on signatures, heuristics, and local behavioral analysis. It generates detection logs on the host itself, but it does not collect or correlate logs from other systems. Its view is limited to the device where it is installed and does not cover the infrastructure as a whole. Although it can send alerts to a SIEM, it does not itself perform the multi-source correlation function described in the question.

2) B — Implement the corresponding solution: SIEM aggregates + correlates; without logs none of this exists....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): SIEM (Security Information and Event Management) aggregates and correlates logs from various sources to detect and respond to incidents. Without logs, SIEM cannot function, highlighting the importance of comprehensive logging.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual

accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

3) C — Implement the corresponding solution: Correlation: VPN login + firewall denied + EDR alert = possible attack.

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): Correlation: VPN login + firewall denied + EDR alert = possible attack. Correlating events from multiple sources (SIEM) improves detection of complex threats.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) D — Implement the corresponding solution: 'combines logs from multiple sources and correlates'=SIEM....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): 'combines logs from multiple sources and correlates'=SIEM.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) D — Implement the corresponding solution: 'VPN login event 8:10, firewall denied 8:11, EDR blocks 8:12' = SIEM correlation.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): 'VPN login event 8:10, firewall denied 8:11, EDR blocks 8:12' = SIEM correlation. SIEM correlation aggregates and analyzes log data from multiple sources to detect complex attack patterns that individual events might miss.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual

accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) D — Implement the corresponding solution: SIEM aggregates + correlates; without logs none of this exists....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): SIEM aggregates + correlates; without logs none of this exists.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) C — Implement the corresponding solution: Correlation: VPN login + firewall denied + EDR alert = possible attack....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): Correlation: VPN login + firewall denied + EDR alert = possible attack. This describes the process of correlating events from multiple sources (VPN authentication logs, firewall deny logs, and EDR alerts) to identify a potential security incident. Security monitoring relies on log correlation and SIEM to detect patterns that individually may not be suspicious but together indicate an attack.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: 'combines logs from multiple sources and correlates' = SIEM....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): 'combines logs from multiple sources and correlates' = SIEM. A Security Information and Event Management (SIEM) system aggregates log data from multiple sources (firewalls, servers, endpoints, applications) and correlates events to detect patterns indicative of security incidents, enabling centralized monitoring and alerting.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) C — Implement the corresponding solution: 'VPN login event at 8:10, firewall denied at 8:11, EDR blocks at 8:12' = SIEM correlation.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.4 (Security Monitoring): 'VPN login event at 8:10, firewall denied at 8:11, EDR blocks at 8:12' = SIEM correlation. This scenario illustrates how a SIEM aggregates and correlates logs from multiple sources (VPN, firewall, EDR) to detect a coordinated attack sequence, enabling timely incident response.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) B — Implement the corresponding solution: SIEM aggregates + correlates; without logs none of this exists....

✓ **Why it is correct:** According to the official CompTIA Security+ exam objectives (SY0-701) for lesson 4.4 (Security Monitoring): SIEM aggregates + correlates; without logs none of this exists. A SIEM (Security Information and Event Management) system collects and correlates log data from multiple sources to detect and respond to security incidents; without proper logging, the SIEM has no data to analyze.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

